

OrientationJ — resources and primary references

Companion appendix to the annotated reference table OrientationJ_References.xlsx. OrientationJ is a series of ImageJ/Fiji plugins by Daniel Sage (Biomedical Imaging Group, EPFL) that characterize local orientation and coherency from the gradient structure tensor. Compiled with AI assistance through targeted web searches of forum.image.sc, GitHub/GitLab, bigwww.epfl.ch, PyPI/bioconda and publisher databases; DOIs cross-checked against PubMed and the publishers. Last updated 7 August 2026.

Appendix A — Primary references of OrientationJ

The publications the plugin authors ask users to cite, as listed on the official page and in the GitHub README:

- Püspöki Z, Storath M, Sage D, Unser M (2016) Transforms and Operators for Directional Bioimage Analysis: A Survey. *Advances in Anatomy, Embryology and Cell Biology*, vol. 219 (Focus on Bio-Image Informatics), Springer.
Main reference on the method — the citation requested by the plugin authors for OrientationJ itself.
[doi:10.1007/978-3-319-28549-8_3](https://doi.org/10.1007/978-3-319-28549-8_3)
- Rezakhaniha R, Agianniotis A, Schrauwen JTC, Griffa A, Sage D, Bouten CVC, van de Vosse FN, Unser M, Stergiopoulos N (2012) Experimental Investigation of Collagen Waviness and Orientation in the Arterial Adventitia Using Confocal Laser Scanning Microscopy. *Biomechanics and Modeling in Mechanobiology* 11(3-4): 461-473.
Reference on the angular distribution; the paper most frequently cited alongside OrientationJ in the companion table.
[doi:10.1007/s10237-011-0325-z](https://doi.org/10.1007/s10237-011-0325-z)
- Fonck E, Feigl GG, Fasel J, Sage D, Unser M, Rüfenacht DA, Stergiopoulos N (2009) Effect of Aging on Elastin Functionality in Human Cerebral Arteries. *Stroke* 40(7): 2552-2556.
Reference on the local (ROI) orientation and coherency measurements.
[doi:10.1161/STROKEAHA.108.528091](https://doi.org/10.1161/STROKEAHA.108.528091)
- Unser M, Sage D, Van De Ville D (2009) Multiresolution Monogenic Signal Analysis Using the Riesz-Laplace Wavelet Transform. *IEEE Transactions on Image Processing* 18(11): 2402-2418.
Reference on the wavelet-based monogenic analysis (MonogenicJ, bundled with OrientationJ from version 2.0.7).
[doi:10.1109/TIP.2009.2027628](https://doi.org/10.1109/TIP.2009.2027628)

Appendix B — Official software & web resources

- OrientationJ official page — Biomedical Imaging Group, EPFL
Official distribution page by Daniel Sage (current version 2.0.7, 21 October 2024): download of OrientationJ.jar, functional description of all modes, recordable ImageJ-macro examples, test images (chirp, artificial fibers, collagen MIP) and a gallery of application cases quoting published usage.
<https://bigwww.epfl.ch/demo/orientationj/>
- Theoretical background (PDF)
Full mathematical description of the gradient structure tensor, orientation, energy and coherency computations underlying every OrientationJ mode.
<https://bigwww.epfl.ch/demo/orientationj/theoretical-background.pdf>
- Source code on GitHub — Biomedical-Imaging-Group/OrientationJ
Open-source Java code (GPL-3.0). The plugins.config lists the seven menu commands: OrientationJ Analysis, Distribution, Directions, Measure, Corner Harris, Dominant Direction and Vector Field. The README restates the four primary references.
<https://github.com/Biomedical-Imaging-Group/OrientationJ>

■ **MonogenicJ** — companion plugin

Wavelet-based multiresolution monogenic analysis of 2D images (orientation, coherency, wavenumber per scale); included in OrientationJ from version 2.0.7.

<https://bigwww.epfl.ch/demo/monogenicj/>

■ **OrientationPy** — EPFL Center for Imaging

The official “pythonic successor” to OrientationJ: a Python library computing greylevel orientations of 2D images and 3D volumes with several gradient techniques. Documentation and example gallery at epfl-center-for-imaging.gitlab.io/orientationpy; installable from PyPI and bioconda.

<https://gitlab.com/epfl-center-for-imaging/orientationpy>

■ **BIII** — BioImage Informatics Index entry

Curated registry entry for OrientationJ (function, platform, licence, links) in the NEUBIAS/BioImage-Analysis tools index.

<https://biii.eu/orientationj>

Appendix C — Community discussions (forum.image.sc)

Selected threads on the Scientific Community Image Forum where OrientationJ usage, parameters and interpretation are discussed — several answered with input from the plugin author:

■ **Coherency in OrientationJ (2023)**

Interpreting coherency values of collagen fibers in second-harmonic-generation images (gel-only control vs. cell-containing samples).

<https://forum.image.sc/t/coherency-in-orientation-j/83176>

■ **OrientationJ dealing with dark regions (2021)**

How dark/background regions influence coherency and the dominant direction; question addressed directly to the plugin author (@daniel.sage).

<https://forum.image.sc/t/orientationj-dealing-with-dark-regions/61179>

■ **How is the “Dominant Direction” calculated? (2020)**

Definition of the orientation and coherency values returned by the Dominant Direction tool and the role of the local-window parameter.

<https://forum.image.sc/t/how-is-the-dominant-direction-calculated/38330>

■ **OrientationJ and angle histograms (2026)**

Why the Distribution histogram can show a large spike at 0° on a test image, and the pre-processing needed to get meaningful orientation histograms.

<https://forum.image.sc/t/orientationj-and-angle-histograms/119187>

■ **OrientationJ and its interpretation for electrospun nanofibers (2017)**

What the energy, orientation and coherency outputs mean when quantifying nanofiber alignment — one of the most common applications in the companion table.

<https://forum.image.sc/t/orientation-j-and-its-interpretation-for-electrospun-nanofibers/3821>

■ **OrientationJ documentation (2022)**

Where to find detailed documentation beyond the official web page (functions of the library, parameters).

<https://forum.image.sc/t/orientationj-documentation/69672>

■ **Macro for plug-in OrientationJ (2017)**

Batch-processing OrientationJ from the ImageJ macro language and saving the result tables automatically.

<https://forum.image.sc/t/macro-for-plug-in-orientationj/7224>

Appendix D — Validation, comparison & related tools

- Morrill EE, Tulepbergenov AN, Stender CJ, Lamichhane R, Brown RJ, Lujan TJ (2016) — FiberFit
“A validated software application to measure fiber organization in soft tissue”, Biomech Model Mechanobiol 15:1467–1478. Accuracy comparison of FFT vs. structure-tensor fiber-orientation methods; listed on the official OrientationJ page as the validation reference for fiber orientation.

<https://doi.org/10.1007/s10237-016-0776-3>

- Hotelling NA, Bharti K, Kriel H, Simon CG (2015) — DiameterJ
“DiameterJ: A validated open source nanofiber diameter measurement tool”, Biomaterials 61:327–338. The fiber-diameter companion plugin that many papers in the table use together with OrientationJ (e.g. entries 13, 86, 87, 104, 194, 293, 296).

<https://doi.org/10.1016/j.biomaterials.2015.02.015>

- Fiji “Directionality” plugin
Fourier-components / local-gradient alternative for orientation histograms, frequently used alongside or compared with OrientationJ (e.g. entries 65, 141, 183, 280, 298).

<https://imagej.net/plugins/directionality>

- Muñoz A, Docaj A, Fernandez J, Carriero A (2025) — FiberO
“FiberO for an automated quantitative analysis of fibers orientation and organization in biological fibrous tissues”, Front Bioeng Biotechnol. Recent open-source tool benchmarked against OrientationJ, Directionality, FiberFit and CT-FIRE.

<https://doi.org/10.3389/fbioe.2024.1497837>

- scikit-image structure tensor (Python)
skimage.feature.structure_tensor — a minimal Python structure-tensor implementation (finite-difference gradients only), cited by the OrientationPy documentation as the closest alternative in the Python ecosystem.

https://scikit-image.org/docs/stable/api/skimage.feature.html#skimage.feature.structure_tensor